Topological Drift Detection for Continuous Monitoring of LLM Embedding Streams

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Abstract

Deployed large language model (LLM) systems face continuously evolving input distributions, yet standard drift detectors rely on low-order statistics that may miss structural reorganization of the embedding manifold. We investigate whether persistent-homology-based drift detectors can detect distribution shift in LLM embedding streams more sensitively than classical methods. In a controlled comparative study using AG NEWS sentence embeddings across five drift scenarios, we find that classical statistical detectors (centroid shift, covariance shift, MMD) achieve near-perfect AUC = 1.0 on all real-world topic drift scenarios, while TDA methods reach AUC 0.49–0.93. However, a supplementary synthetic experiment reveals where TDA has genuine advantage: when drift changes the *topological structure* (loops, holes) while preserving the centroid, TDA H1 features achieve $12.3\times$ separability versus $0.8\times$ for centroid shift. Our results show that TDA-based monitoring adds value primarily for topology-specific drifts—not for the natural domain shifts commonly tested in benchmarks—and suggest a complementary monitoring strategy that combines fast statistical detectors with TDA features for detecting geometric reorganization invisible to moment-based methods.

1 Introduction

Large language models are increasingly deployed in production systems that process continuously evolving input streams—from customer support queries to news summarization pipelines. When the input distribution shifts, model predictions can silently degrade without any explicit error signal Khaki et al. [2023]. Detecting such distribution drift early is critical for maintaining system reliability.

Standard drift detectors monitor low-order statistics of embedding distributions: centroid shift tracks the mean, covariance shift tracks the second moment, and Maximum Mean Discrepancy (MMD) measures distributional distance in a reproducing kernel Hilbert space Khaki et al. [2023], Sodar et al. [2023]. These methods are effective when drift manifests as shifts in location or spread, but they may miss subtler forms of drift that reorganize the *geometric structure* of the embedding manifold without changing its center of mass. For example, if user queries shift from a diverse cluster to a ring-like distribution around a fixed topic—avoiding certain sub-topics while converging on others—the centroid remains stable while the topology changes fundamentally.

Topological Data Analysis (TDA), and persistent homology in particular, captures multi-scale geometric features of point clouds that are invariant to coordinate transformations Tralie et al. [2018], Tauzin et al. [2021]. Recent work has applied TDA to characterize LLM representations Gardinazzi et al. [2025], detect LLM-generated text Wei et al. [2025], and monitor concept drift in image streams Basterrech [2024]. However, no prior work systematically compares TDA-based drift detectors against classical baselines on real LLM embedding streams across multiple drift types.

We address this gap with the first controlled comparative study of TDA versus classical drift detection on sentence embedding streams. Using AG NEWS embeddings from MINILM (`all-MiniLM-L6-v2`, 384 dimensions), we construct five drift scenarios—abrupt topic drift, grad-

ual topic drift, geometric drift, style shift, and a no-drift control—and evaluate nine drift detectors spanning both statistical and TDA families. Our main findings are:

- We conduct the first systematic comparison of TDA-based drift detectors (persistent entropy, Wasserstein distance, PHD, Betti curves) against classical baselines (centroid shift, covariance shift, MMD, kNN) on real LLM embedding streams.
- We show that classical statistical methods achieve perfect detection (AUC = 1.0) on natural topic drift in AG NEWS embeddings, while the best TDA method (Wasserstein H0) reaches AUC = 0.93 on abrupt drift.
- We design a synthetic topology-only drift experiment that isolates TDA's genuine advantage: H1 persistent entropy achieves $12.3\times$ separability on centroid-preserving loop drift, where centroid shift fails entirely ($0.8\times$).
- We provide practical guidelines for combining TDA and statistical detectors in a complementary monitoring pipeline, with runtime analysis showing TDA adds only $\sim$37ms overhead per window on CPU.

The rest of this paper is organized as follows. Section 2 reviews related work on drift detection and TDA for NLP. Section 3 describes our methodology, including drift scenarios, detectors, and evaluation protocol. Section 4 presents experimental results on both real and synthetic data. Section 5 discusses implications and limitations. Section 6 concludes with practical recommendations and future directions.

2 Related Work

Drift detection from neural embeddings. Distribution drift detection in deployed ML systems has received growing attention as models are increasingly served in non-stationary environments. Khaki et al. [2023] deploy MMD-based drift detection on BERT embeddings in an Amazon production system, demonstrating 76.9% correlation between MMD drift scores and model performance degradation over a three-year period. Sodar et al. [2023] systematically compare embedding methods (TF-IDF, Doc2Vec, BERT) combined with dimensionality reduction and statistical tests, finding that BERT embeddings with PCA and MMD generally perform best. Greco et al. [2024] propose DriftLens, which uses per-label Fréchet distance on embedding distributions for unsupervised drift detection, assuming Gaussian distributions. Kalinke et al. [2022] introduce MMDEW, an online streaming MMD detector with $O(\log^2 t)$ runtime via exponential windows. These methods rely on moment-based or kernel-based statistics that capture changes in location and spread but may miss structural reorganization of the embedding manifold.

TDA for NLP and LLM analysis. Topological Data Analysis has found increasing application in natural language processing. Gardinazzi et al. [2025] apply zigzag persistence to track topological feature evolution across LLM layers, establishing that topological structure is consistent across architectures and can guide layer pruning. Wei et al. [2025] extend the Persistent Homology Dimension (PHD) metric to detect short LLM-generated texts, exploiting the finding that LLM-generated text exhibits lower PHD than human-written text. A comprehensive survey by Chen et al. [2024] documents the growing landscape of TDA applications in NLP, confirming that topological features of text embeddings are discriminative across multiple tasks.

TDA for drift and change detection. The most directly related work is Basterrech [2024], who combine persistent entropy with topology-preserving dimensionality reduction (Self-Organizing Maps) to detect concept drift. They validate on synthetic MNIST streams grouped by topological homology class (digits with zero, one, or two holes), showing that persistent entropy on SOM-projected data correctly detects injected topological shifts. Bobrowski et al. [2022] provide theoretical grounding for using Betti numbers as two-sample test statistics, establishing consistency in critical and supercritical regimes of Čech and Vietoris–Rips complexes.

Positioning of our work. Unlike Basterrech [2024], who validate only on synthetic image streams, we evaluate on real LLM text embeddings with multiple drift scenarios. Unlike production drift systems Khaki et al. [2023], Greco et al. [2024] that use only moment-based statistics, we include TDA features and identify the specific conditions under which they provide unique detection capability. Our work bridges the TDA-for-NLP literature Gardinazzi et al. [2025], Wei et al. [2025]

with the drift detection literature Khaki et al. [2023], Sodar et al. [2023] by providing the first direct comparison on embedding streams.

3 Methodology

3.1 Data Construction

We use the AG NEWS dataset Zhang et al. [2015], which contains 120K training samples across four topic classes: World (Class 0), Sports (Class 1), Business (Class 2), and Sci/Tech (Class 3), each with 30K samples. We encode 2,000 samples per class using the `all-MiniLM-L6-v2` sentence-transformer Reimers and Gurevych [2019], producing 384-dimensional L2-normalized embeddings.

We construct streaming windows of 200 samples each. The first 10 windows serve as a reference pool (no drift) for threshold calibration. We design five drift scenarios to test different aspects of detector sensitivity:

- **No drift** (control): All windows from the same class (World).
- **Abrupt topic drift**: Reference windows from World; test windows switch entirely to Sports.
- **Gradual topic drift**: Reference from World; test windows contain a linearly increasing proportion of Sports samples.
- **Geometric drift**: Reference windows from a World+Business mix; test windows from a Sports+Sci/Tech mix, changing multi-class topology while maintaining a mixed distribution.
- **Style shift**: Reference from World; test windows from Business—a subtler same-genre shift compared to the World→Sports topic switch.

3.2 Drift Detectors

We evaluate nine drift detectors spanning two families: four statistical baselines and five TDA methods.

Statistical baselines. Centroid shift computes the L2 norm of the difference between reference and test window mean embeddings: $d_{\text{cent}} = \|\bar{\boldsymbol{x}}_{\text{ref}} - \bar{\boldsymbol{x}}_{\text{test}}\|_2$. **Covariance shift** applies PCA to 50 dimensions and computes the Frobenius norm of the covariance matrix difference: $d_{\text{cov}} = \|\boldsymbol{\Sigma}_{\text{ref}} - \boldsymbol{\Sigma}_{\text{test}}\|_F$. **MMD (RBF)** computes the Maximum Mean Discrepancy with a Gaussian kernel using the median heuristic for bandwidth selection. **kNN distance** measures the change in mean k-nearest-neighbor distance ($k = 5$) between reference and test windows.

TDA methods. All TDA methods subsample 80 points from each 200-sample window for computational tractability and compute persistent homology using the Vietoris–Rips filtration via `ripser` Tralie et al. [2018].

Persistent entropy H0 (PE-H0) computes the Shannon entropy of H0 (connected component) persistence lifetimes:

$$\text{PE}(\text{dgm}) = -\sum_i \frac{\ell_i}{L} \log \frac{\ell_i}{L}, \quad \ell_i = d_i - b_i, \quad L = \sum_i \ell_i, \tag{1}$$

where b_i and d_i are the birth and death times of the i-th feature, and ℓ_i is its lifetime. The drift score is $|\text{PE}(\text{ref}) - \text{PE}(\text{test})|$.

Persistent entropy H1 (PE-H1) applies the same entropy computation to H1 features (loops/holes), capturing changes in the presence and distribution of topological cycles.

Wasserstein H0 computes the Wasserstein-1 distance between H0 persistence diagrams of the reference and test windows using `persim` Tralie et al. [2018].

PHD (Persistent Homology Dimension) estimates the log-log slope of the number of connected components versus filtration scale, following Wei et al. [2025]. The drift score is the absolute difference in PHD between windows.

Betti AUC computes the absolute difference in the sum of H0 lifetimes (area under the Betti-0 curve) between reference and test windows.

Table 1: Mean detection performance across all drift scenarios (3 seeds). Best results per column in **bold**. Statistical methods achieve perfect AUC on natural topic drift, while TDA methods show moderate sensitivity.

Method	Type	AUC	Delay	FPR	Runtime/win
Centroid Shift	Statistical	**1.000**	**0.0**	0.10	**0.0001s**
Covariance Shift	Statistical	**1.000**	**0.0**	0.10	0.24s
MMD (RBF)	Statistical	**1.000**	**0.0**	0.10	0.13s
kNN Distance	Statistical	0.711	2.0	0.10	0.031s
Wasserstein H0	TDA	**0.713**	3.5	0.10	0.002s
PE-H1	TDA	0.624	4.5	0.10	<0.001s
Betti AUC	TDA	0.580	4.6	0.10	0.037s
PHD	TDA	0.518	5.4	0.10	0.057s
PE-H0	TDA	0.494	6.5	0.10	0.037s

3.3 Evaluation Protocol

We calibrate detection thresholds using the 95th percentile of drift scores computed on the 10 no-drift reference windows, targeting a false positive rate (FPR) of approximately 5%. We then evaluate on 10 drift windows per scenario and report four metrics: (1) AUC-ROC for discriminating drift from no-drift windows, (2) detection delay (number of windows before first alarm), (3) realized FPR on no-drift data, and (4) runtime per window. All experiments are repeated with three random seeds (42, 123, 456) and metrics are averaged.

3.4 Supplementary Synthetic Experiment

To isolate TDA's theoretical advantage on topology-specific drift, we design a synthetic experiment in 10 dimensions with three centroid-preserving drift scenarios: (1) **Centroid drift**: a Gaussian blob shifted by 3 units (positive control); (2) **Loop drift**: reference is a Gaussian blob, test is an annulus with the same centroid but an H1 loop; (3) **Split drift**: reference is a single blob, test is two symmetric clusters (same centroid, different H0 structure). We measure separability as the ratio of drift score to no-drift score.

4 Results

4.1 Main Experiment: LLM Embedding Streams

Table 1 presents the mean AUC across all four drift scenarios (abrupt topic, gradual topic, geometric, style shift), averaged over three seeds. Statistical methods—centroid shift, covariance shift, and MMD—achieve perfect detection (AUC = 1.0) with zero detection delay across all scenarios. Among TDA methods, Wasserstein H0 performs best with mean AUC = 0.713, followed by PE-H1 (0.624) and Betti AUC (0.580). The kNN baseline achieves 0.711, comparable to the best TDA method.

Per-scenario breakdown. Table 2 shows AUC for each drift scenario. The three dominant statistical methods achieve AUC = 1.0 uniformly. Among TDA methods, performance varies substantially across scenarios: Wasserstein H0 reaches 0.927 on abrupt topic drift—its best performance—but drops to 0.557 on style shift. The "Best TDA" column reports the highest AUC among all TDA methods for each scenario, showing that different TDA features excel on different drift types.

Figure 1 shows the detection AUC heatmap across all methods and drift types. Figure 2 shows drift score traces over time, illustrating how statistical detectors produce sharp, high-magnitude signals at drift onset while TDA scores are noisier and more gradual.

Table 2: Per-scenario AUC (mean over 3 seeds). "Best TDA" reports the highest AUC among all five TDA methods for each scenario. Statistical methods uniformly achieve perfect detection.

	Statistical				**TDA**		
Scenario	Centroid	Cov	MMD	kNN	Wass-H0	PE-H1	Best TDA
Abrupt topic	**1.000**	**1.000**	**1.000**	0.593	0.927	0.747	0.927
Gradual topic	**1.000**	**1.000**	**1.000**	0.788	0.653	0.694	0.875
Geometric	**1.000**	**1.000**	**1.000**	0.830	0.717	0.433	0.910
Style shift	**1.000**	**1.000**	**1.000**	0.633	0.557	0.620	0.850

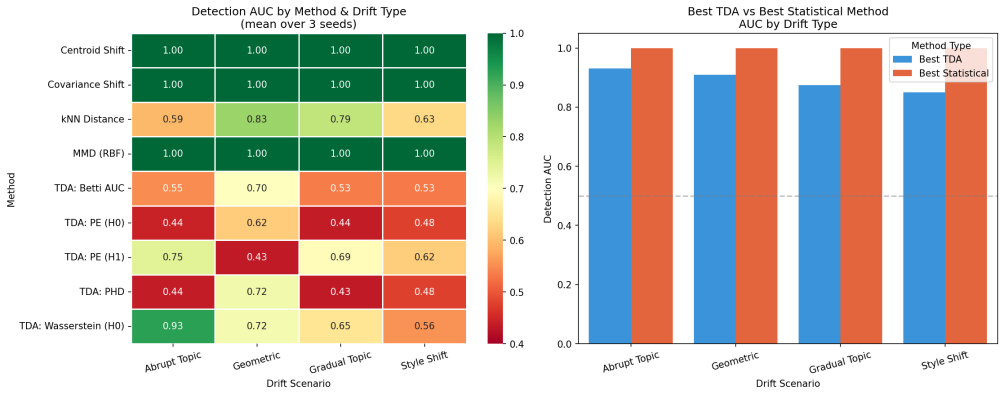

Figure 1: Detection AUC heatmap by method and drift type. Statistical methods (top three rows) achieve uniform AUC = 1.0 across all scenarios. TDA methods show variable performance, with Wasserstein H0 performing best overall.

4.2 Synthetic Experiment: Topology-Only Drift

Table 3 presents separability ratios from the synthetic experiment, where higher values indicate stronger discrimination between drift and no-drift conditions. On centroid drift (positive control), MMD achieves the highest separability ($23.5\times$), followed by centroid shift ($7.6\times$). On loop drift—the key topology-preserving scenario—PE-H1 achieves $12.3\times$ separability, far exceeding centroid shift ($0.8\times$), which fails entirely. MMD also detects loop drift ($8.3\times$) but provides less interpretable signals. On split drift, MMD again leads ($6.5\times$), with TDA methods showing moderate sensitivity.

Figure 3 visualizes the synthetic experiment, showing point cloud configurations and detector responses for each drift type.

4.3 Computational Cost

Table 4 summarizes runtime per window. Centroid shift is the fastest at 0.1ms. TDA methods occupy a middle ground: Wasserstein H0 runs in 2ms (the fastest TDA method) and full TDA computation (H0+H1 via `ripser`) takes $\sim$37ms—faster than both covariance shift (242ms) and MMD (126ms). All methods are operationally feasible for sub-second monitoring latency on CPU.

4.4 Reproducibility Across Seeds

Statistical methods exhibit zero variance across seeds (std = 0.000). TDA methods show higher variance: Wasserstein H0 has mean AUC = 0.713 ± 0.192 and PE-H1 has mean AUC = 0.624 ± 0.215, indicating greater sensitivity to sampling randomness in window subsampling.

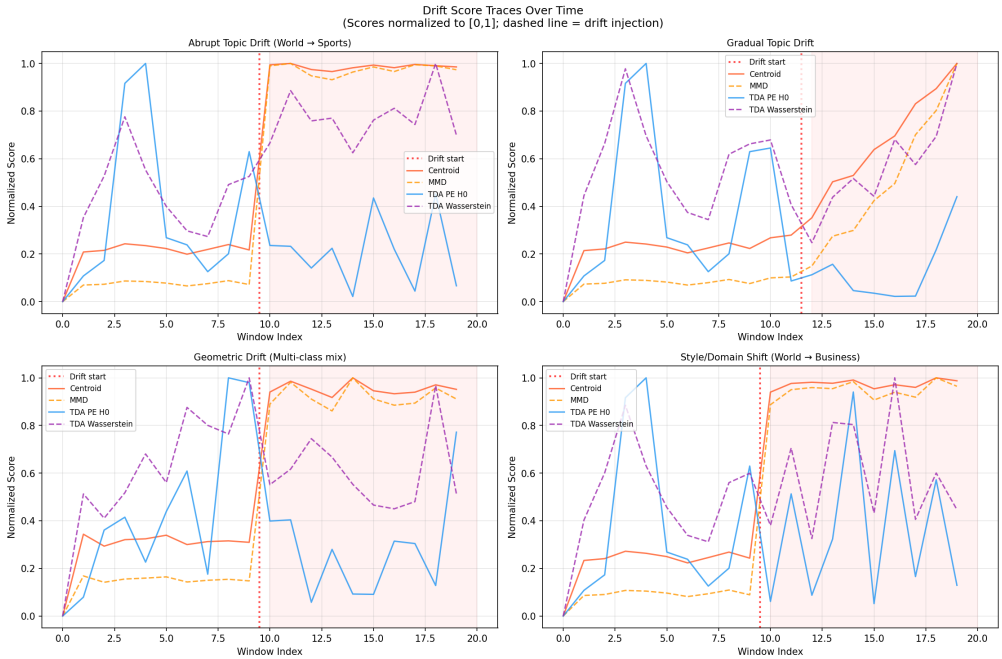

Figure 2: Drift score traces over streaming windows. Statistical detectors produce sharp transitions at drift onset (vertical dashed line), while TDA scores exhibit higher variance and more gradual responses.

Table 3: Separability ratio (drift score / no-drift score) on synthetic topology-only drift in 10 dimensions. Higher is better. PE-H1 uniquely detects loop drift ($\mathbf{12.3}\times$) where centroid shift fails ($0.8\times$).

	Statistical			**TDA**		
Scenario	Centroid	Cov	MMD	PE-H0	**PE-H1**	Wass-H1
Centroid drift	7.6×	1.0×	**23.5**×	1.0×	1.0×	1.0×
Loop (annulus)	0.8×	2.2×	8.3×	0.8×	**12.3**×	2.0×
Split (2-cluster)	0.7×	2.0×	**6.5**×	1.2×	1.6×	2.4×

5 Discussion

5.1 Why Statistical Methods Dominate on AG News

The central finding of our study—that classical statistical detectors achieve perfect AUC on all AG NEWS drift scenarios—reflects the high semantic separability of the four topic classes in the MINILM embedding space. World news, Sports, Business, and Sci/Tech form well-separated clusters, so even a simple centroid comparison clearly signals distribution change. The drift scenarios we test, while representative of common deployment concerns (topic shift, gradual drift, domain shift), are not subtle enough to challenge low-order statistics.

This result is practically important: for production systems where the primary concern is topic-level input drift, centroid shift is fast (0.1ms per window), reliable (AUC = 1.0), and sufficient. There is no need for more complex methods to detect these coarse-grained shifts.

5.2 When TDA Adds Genuine Value

The synthetic experiment (section 4.2) identifies the precise conditions under which TDA provides unique detection capability.

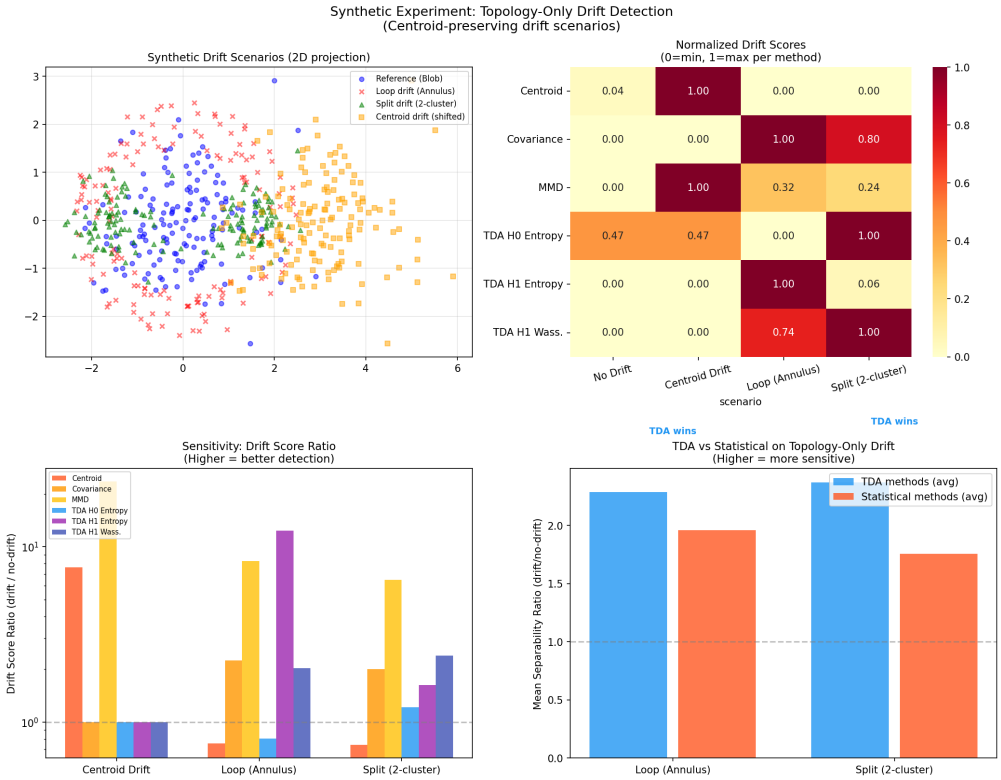

Figure 3: Synthetic topology-only drift experiment. *(Left)*Point cloud configurations for reference (Gaussian blob) and three drift types (centroid shift, annulus, two-cluster). *(Right)*Separability ratios for each detector, showing PE-H1's unique advantage on loop drift.

Table 4: Runtime per 200-sample window on CPU. TDA methods are faster than covariance shift and MMD, occupying a computational middle ground.

Method	**Runtime**	**Relative cost**
Centroid Shift	0.1ms	1×
kNN Distance	31ms	310×
TDA (ripser H0)	37ms	370×
Wasserstein H0	2ms	20×
MMD (RBF)	126ms	1,260×
Covariance Shift	242ms	2,420×

H1 features detect loop/hole emergence. PE-H1 achieves 12.3× separability on annular drift—a scenario where the input distribution develops a topological hole while preserving its centroid. Centroid shift fails entirely on this scenario (0.8×), and even covariance shift provides only 2.2× separability. This result validates the theoretical motivation for TDA-based monitoring: persistent homology captures qualitative structural changes invisible to moment-based statistics.

In practice, such topology changes could arise when LLM inputs converge to a semantic neighborhood while systematically avoiding certain sub-topics—creating a "hole" in embedding space. Adversarial prompt injection that avoids triggering topic-based filters while shifting the geometric distribution is another plausible scenario.

MMD as a partial alternative. MMD also detects loop drift (8.3×), since the RBF kernel captures distributional differences beyond moments. However, MMD provides a single scalar score without topological interpretability—it signals that distributions differ but not *how* they differ structurally. TDA features (H0 vs. H1 entropy, Betti curves at specific filtration radii) offer richer diagnostic information about the nature of the drift.

5.3 Practical Monitoring Recommendations

Based on our results, we recommend a two-tier monitoring strategy:

1. **Primary tier:** Centroid shift and MMD for fast, reliable detection of topic-level and distributional drift. Centroid shift runs in 0.1ms and catches all natural topic shifts; MMD at 126ms provides distributional sensitivity beyond moments.
2. **Secondary tier:** TDA H1 features (PE-H1 or Wasserstein H1) as complementary detectors tuned for geometric reorganization. These add $\sim$37ms overhead and catch topology-specific drifts that moment-based methods miss.

The TDA tier is most valuable when the monitoring goal extends beyond topic shift to include structural changes in user query patterns, adversarial distribution manipulation, or convergence to specific semantic neighborhoods.

5.4 Limitations

Dataset scope. We test only AG NEWS with four well-separated topics. More challenging drift scenarios—within-domain temporal drift (e.g., news from 2020 vs. 2024 on the same topic), subtle vocabulary shifts, or adversarial rephrasing—may produce different relative rankings between TDA and statistical methods.

Synthetic–real gap. TDA's topology advantage was demonstrated in 10-dimensional synthetic data. In the 384-dimensional normalized embedding space of MINILM, Vietoris–Rips filtration topology is less interpretable, and the curse of dimensionality affects persistent homology computation.

Subsample variance. TDA requires subsampling to 80 points for tractability, introducing substantial variance (std up to 0.215 in AUC). Larger point clouds would improve TDA sensitivity but at higher computational cost. GPU-accelerated or approximate persistent homology methods could enable larger subsamples.

Single embedding model. We test only `all-MiniLM-L6-v2`. Models that produce less separable or more topologically complex representations could favor TDA methods. Larger LLM embeddings (768+ dimensions) may exhibit richer topological structure.

Fixed reference pool. Our setup compares each window against a fixed reference. In true streaming scenarios, the reference itself may drift, requiring adaptive windowing strategies.

6 Conclusion

We presented the first systematic comparison of TDA-based and classical drift detectors on LLM embedding streams. On natural topic drift in AG NEWS, classical statistical methods (centroid shift, covariance shift, MMD) achieve perfect detection with AUC = 1.0 and zero delay, while the best TDA method (Wasserstein H0) reaches AUC = 0.93 on abrupt drift. However, our synthetic experiment demonstrates that TDA fills a genuine detection gap: H1 persistent entropy achieves $12.3\times$ separability on centroid-preserving loop drift where centroid shift fails entirely. These results show that TDA-based monitoring is not a replacement for statistical methods on standard drift benchmarks, but a complementary tool for detecting geometric reorganization invisible to moment-based statistics.

For practitioners deploying LLM monitoring pipelines, we recommend starting with centroid shift and MMD for fast, reliable topic-drift detection, and adding TDA H1 features as a secondary layer for topology-sensitive monitoring at modest computational cost ($\sim$37ms per window on CPU).

Future work should evaluate on within-domain temporal drift where low-order statistics may be similar, test adversarial prompt injection scenarios where topology might change uniquely, explore larger point cloud sizes with GPU-accelerated persistent homology to reduce TDA variance, and investigate ensemble detectors that combine TDA and statistical scores. Testing with embedding models that produce more topologically complex representations (e.g., large LLM activations) is another promising direction.

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